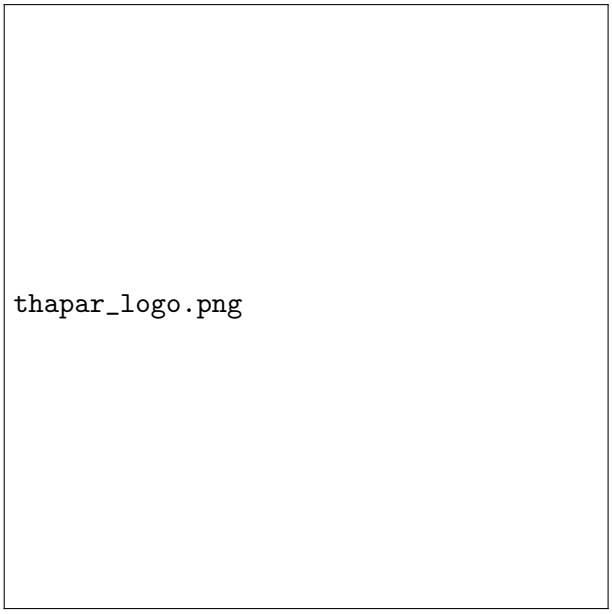

UniCore: The Unified Campus Infrastructure

Full-Stack Architecture, High-Concurrency Database Layer & Automated Deployments

Comprehensive Research Proposal

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Abstract

UniCore represents a paradigm shift in institutional infrastructure, evolving beyond a mere database management system into a holistic **Unified Campus Infrastructure**.

This research proposal delineates the end-to-end architectural integrity of UniCore. At its core lies a strictly normalized (BCNF) PostgreSQL cluster across 8 domain schemas. This data layer guarantees absolute ACID transaction safety during catastrophic concurrent enrollment spikes by utilizing row-level locking (`SELECT FOR UPDATE`) and immutable JSON audit ledgers.

Building upon this foundation is a robust Microservices API Backend (Node.js/Express) that acts as the primary gateway, providing secure JWT-based stateless routing. The client interface is driven by a highly dynamic, component-based Frontend built with React, Vite, and Tailwind CSS. This combination delivers a highly responsive, accessible (WCAG 2.1), and visually profound web design tailored for 30,000+ active students.

Finally, the system's operational continuity is maintained via automated CI/CD deployment pipelines utilizing GitHub Actions. Furthermore, this proposal explores the deep ethical implications, algorithmic inclusivity, and rigorous project management frameworks ensuring the timely delivery of this monumental infrastructure.

By targeting a Time-To-Acknowledgement (TTA) of ≤ 2 hours, UniCore aims to completely redefine campus operational efficiency.

Contents

1 Introduction

1.1 Context and Motivation

Modern higher education institutions are vast, dynamic ecosystems comprising tens of thousands of active nodes—students, faculty, administrative staff, and automated physical infrastructure. Thapar Institute of Engineering & Technology (TIET), currently serving over 30,000 active students, is a prime example of an institution that has scaled tremendously. However, the digital infrastructure governing these domains has historically evolved in a piecemeal fashion.

Currently, universities deploy disparate, siloed systems for different operational domains. A student's identity exists independently in the Academic Portal, the Hostel Management System, the Library Circulation Database, and the Financial Ledger. This architectural fragmentation causes severe operational friction. When a student pays their semester fee, the reconciliation between the finance department and the academic registration module often requires batch processing or manual intervention.

1.2 Problem Statement

The primary crux of the issue lies in **Concurrency, Consistency, and Centralization (The 3 Cs)**. During peak rushes—such as the first hour of hostel bed allotment or elective course registration—thousands of concurrent transactions bombard the network. In weakly isolated systems, this results in race conditions. Students are often allocated the same physical hostel bed (the "Double-Booking" anomaly) or enrolled in an elective beyond its maximum capacity.

Furthermore, traditional systems suffer from poor web design and rigid user interfaces. Legacy portals are rarely mobile-responsive, lack modern accessibility standards, and offer a dismal User Experience (UX).

1.3 Reframing the Paradigm: The Unified Campus Infrastructure

UniCore is not simply a database. It is an **Comprehensive Unified Infrastructure**.

- **The Core Database (PostgreSQL):** Manages the fundamental state, concurrency locks, and memory (data) allocation.
- **The System APIs (Node.js Backend):** Exposes system calls to the outside world, validating permissions and managing process routing.
- **The Client Interface (React Frontend):** Provides the interactive, visual interface for the user to interface with the kernel.

Strategic Vision
By treating university domains (Academics, Hostels, Library) as micro-processes sharing a common kernel, UniCore ensures complete harmony between state and execution.

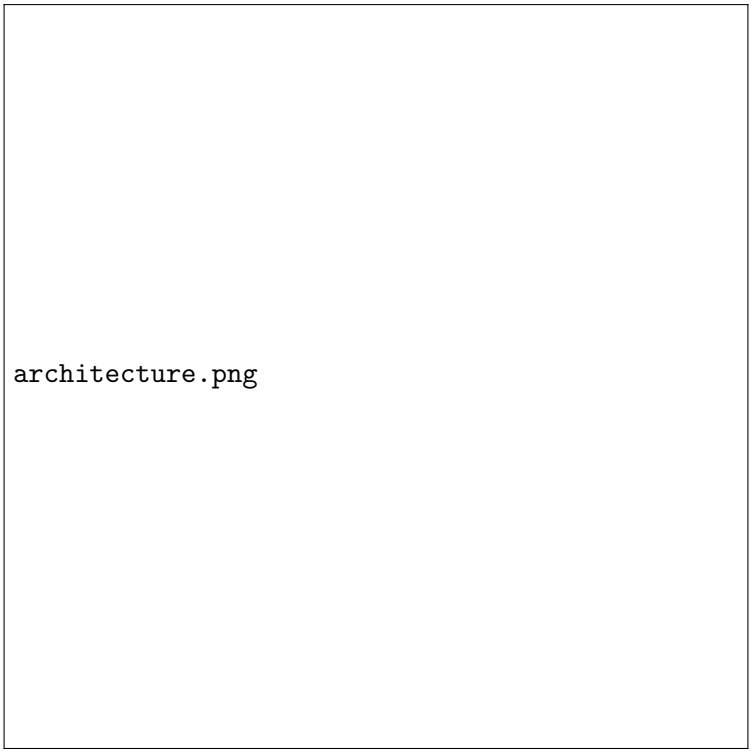

Figure 1: UniCore System Architecture Component Diagram

2 Frontend Architecture & Modern Web Design

The "Client Interface" of UniCore is designed to be as performant as it is visually stunning. We recognize that for 30,000+ students, the frontend is the entirety of their experience with the system.

2.1 Component-Driven React Vite Architecture

The frontend is constructed using **React.js** bootstrapped with **Vite**. Vite provides lightning-fast Hot Module Replacement (HMR) during development and highly optimized Rollup-based bundling for production deployments.

- **Modular Components:** The UI is strictly decomposed into reusable components (e.g., `DataCard`, `RegistrationModal`, `TimelineView`). This mirrors the BCNF normalization of the database, ensuring DRY (Don't Repeat Yourself) principles in the view layer.
- **Client-Side Routing:** Utilizing React Router, UniCore functions as a Single Page Application (SPA), eliminating page reloads and providing a native-app-like fluidity.

2.2 Advanced Styling with Tailwind CSS & PostCSS

To achieve a modern, premium aesthetic, UniCore bypasses legacy CSS frameworks in favor of **Tailwind CSS** processed via **PostCSS**.

- **Utility-First Design:** Rapid prototyping and strict design system adherence are guaranteed. We utilize a customized `tailwind.config.js` to define TIET's specific brand

colors, ensuring institutional consistency.

- **Responsive & Mobile-First:** Every interface, from the complex 7-day academic timetable to the library fine ledger, is engineered to degrade gracefully to mobile viewports.
- **Micro-Animations:** Smooth transitions and hover states are implemented to provide tactile feedback without compromising layout rendering times (Core Web Vitals).

2.3 State Management & Data Fetching

Synchronizing the UI with the backend kernel requires robust state management. UniCore utilizes asynchronous data fetching libraries (like React Query or Redux Toolkit) to cache requests, handle loading states skeleton screens, and smoothly render error boundaries if a transaction aborts.

3 Backend & API Microservices

Figure 2: Authentication & Microservices Flow Diagram

The backend acts as the critical intermediary—the system call handler—between the React shell and the PostgreSQL kernel.

3.1 Node.js and Express Framework

The server infrastructure is built on **Node.js** utilizing the **Express** framework. Node.js's asynchronous, event-driven architecture is ideally suited for handling the high volume of con-

current I/O requests generated during peak campus events (e.g., thousands of simultaneous grade-checks).

3.2 Stateless RESTful Architecture

To ensure horizontal scalability, the backend is entirely stateless.

- **JWT Authentication:** Session state is not stored on the server. Instead, cryptographically signed JSON Web Tokens (JWT) are issued upon login. Every subsequent API request must include this token in the `Authorization` header.
- **Role-Based Access Control (RBAC):** Middleware functions intercept incoming requests and decode the JWT to verify the user's role (Student, Faculty, Admin). A student attempting to POST to the `/api/grades` endpoint will be rejected with a `403 Forbidden` before the request ever reaches the database kernel.

3.3 Rate Limiting and Security Middleware

To protect the system from DDoS attacks or accidental frontend infinite loops, the backend implements strict rate limiting (e.g., max 100 requests per IP per minute). Additionally, tools like `Helmet.js` are used to secure HTTP headers against XSS and clickjacking attacks.

4 Deployment & CI/CD DevOps Pipelines

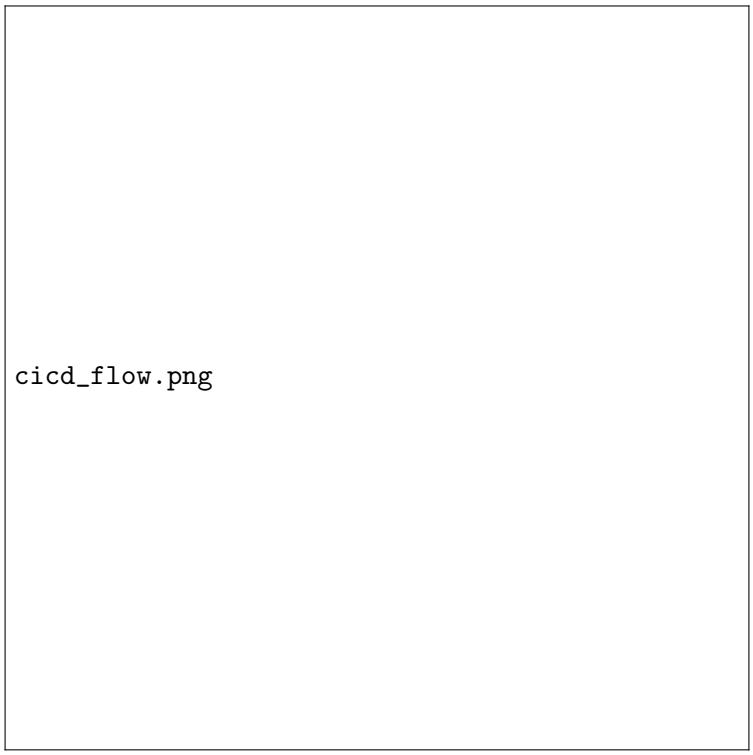

Figure 3: CI/CD Deployment Pipeline via GitHub Actions

A system is only as good as its deployment strategy. UniCore embraces modern DevOps practices to ensure continuous integration and zero-downtime delivery.

4.1 Automated GitHub Actions

As evidenced by the `.github/workflows/deploy.yml` configuration within the repository, UniCore utilizes GitHub Actions for its CI/CD pipeline.

- **Continuous Integration:** Every `git push` triggers an automated workflow that installs dependencies, lints the codebase, and executes unit tests.
- **Automated Build:** The React/Vite frontend is automatically bundled into optimized static assets.
- **Continuous Deployment (GitHub Pages & VPS):** The documentation hub and frontend artifacts are automatically deployed to GitHub Pages, while the backend Node.js services are containerized via Docker and deployed to institutional Virtual Private Servers (VPS).

4.2 Load Balancing and Containerization

By containerizing the Node.js backend using **Docker**, the university IT department can utilize orchestration tools (like Kubernetes or Docker Swarm) to spin up multiple instances of the backend API behind a Load Balancer (e.g., Nginx) during the "Elective Rush," dynamically scaling resources to meet demand.

5 The Core Database: Database Design & Normalization

The cornerstone of UniCore is its schema design. The system encompasses over 35 tables partitioned across 8 distinct domain schemas.

5.1 Domain Schemas

Schema Name	Operational Responsibility
`core`	Global entity definitions (Students, Faculty, Authentication).
`academics`	Courses, Prerequisite mappings, Department structures.
`enrollment`	Student course registrations, Timetables, Attendance.
`hostel`	Residential blocks, Room capacities, Live bed allocations.
`library`	Asset catalogs, Book issues, Fine ledgers, Digital media.
`examinations`	Exam scheduling, Seat matrices, Seating arrangements, Grading.
`finance`	Fee structures, Payment gateways, Dues, Scholarship allocations.
`audit`	Immutable JSONB ledgers and security incident tracking.

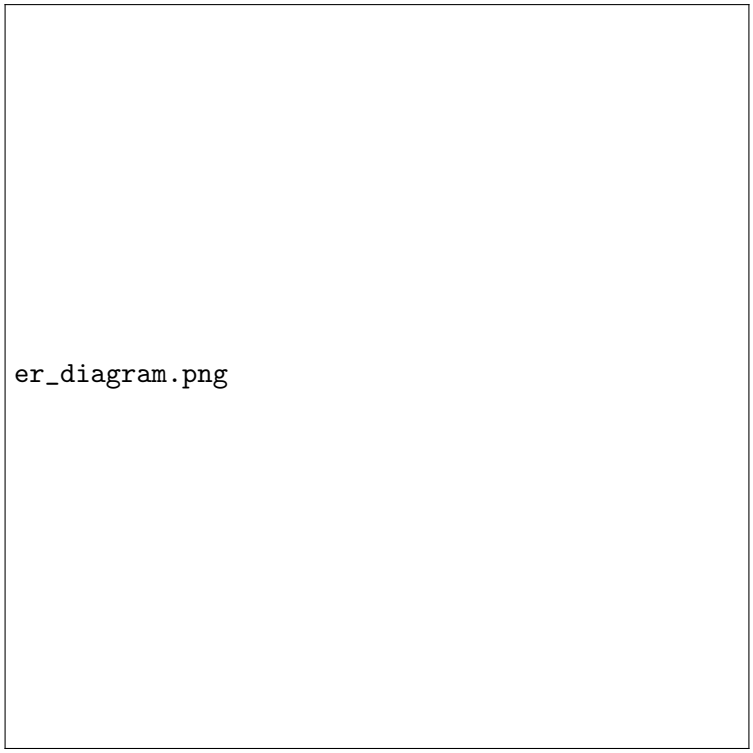

Figure 4: UniCore Normalized ER Diagram (BCNF)

5.2 Boyce-Codd Normal Form (BCNF)

Every relation within UniCore is strictly decomposed into BCNF. A table is in BCNF if and only if for every non-trivial functional dependency $X \rightarrow Y$, X is a superkey. By enforcing BCNF, UniCore guarantees the elimination of update, insertion, and deletion anomalies. For instance, a student's current residential address is stored strictly in `core.students` and accessed via foreign keys across the library and academic schemas. If a student changes their address, the update occurs in exactly one row, preventing data drift.

6 Concurrency Control & Transaction Safety

UniCore is architected to handle "stampede" events—periods of intense concurrent load directed at highly constrained resources.

6.1 Pessimistic Row-Level Locking

For resources like Hostel Beds, where physical constraints dictate absolute exclusivity, UniCore utilizes Pessimistic Locking.

Listing 1: Hostel Allotment Lock Mechanism

```
BEGIN;
-- Find an available bed and lock it exclusively
SELECT bed_id INTO v_allocated_bed
```

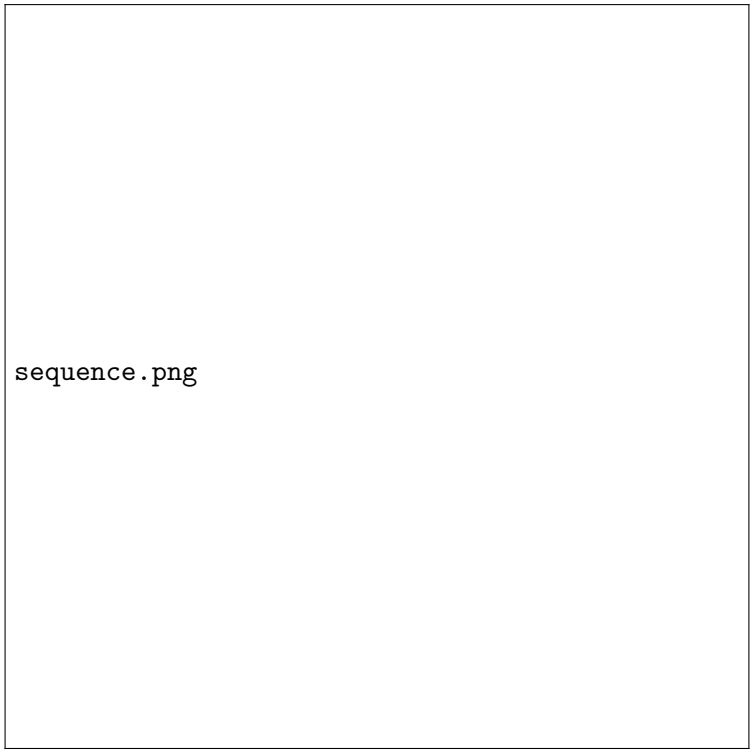

Figure 5: Concurrency Control - Hostel Allotment Sequence Diagram

```
FROM hostel.beds
WHERE block_id = 'A' AND status = 'AVAILABLE'
LIMIT 1
FOR UPDATE SKIP LOCKED;

IF v_allocated_bed IS NOT NULL THEN
    UPDATE hostel.beds
      SET status = 'OCCUPIED'
      WHERE bed_id = v_allocated_bed;

    INSERT INTO hostel.allocations (student_id, bed_id)
      VALUES (p_student_id, v_allocated_bed);
    COMMIT;
ELSE
    ROLLBACK;
    RAISE EXCEPTION 'No available beds in Block A';
END IF;
```

The `SKIP LOCKED` directive ensures that concurrent transactions do not queue behind a locked row; instead, they immediately fetch the next available row, drastically maximizing transaction throughput (TPS).

6.2 Advisory Locks

For operations that do not map directly to a single table row—such as validating overall exam hall capacity before seating a student—UniCore employs `pg_advisory_xact_lock()`. This generates an application-defined lock in the database memory space, ensuring deterministic serialization of abstract business rules.

Scenario	**Isolation Level**	**Mechanism**	**Enforced Guarantee**
Hostel Bed Allotment	`SERIALIZABLE`	`SELECT FOR UPDATE`	Zero double-booking
Exam Seat Registration	`READ COMMITTED`	`pg_advisory_xact_lock()`	Seat count $\leq$ Capacity
Library Book Issue	`READ COMMITTED`	Decrement `total_copies`	Available count ≥ 0
Grade Record Updates	`READ COMMITTED`	`UPSERT`	Updates row without duplication

7 Security, Triggers, and Audit Ledgers

7.1 Automated PL/SQL Triggers

To enforce business rules immutably, UniCore embeds logic directly into the database via Triggers. This guarantees that no faulty application code can ever violate university policy.

- **BEFORE Triggers:** A `BEFORE INSERT` trigger on the `enrollment` table checks the student's fine ledger. If outstanding library fines exceed Rs. 500, the database aborts the transaction, preventing the student from registering for new courses.
- **AFTER Triggers:** Every table possesses an `AFTER UPDATE` trigger that captures the `OLD` and `NEW` pseudo-records, computes a JSON diff, and pushes the event into `audit.transaction_logs`.

7.2 Forensic Auditability

The `audit.transaction_logs` table is an append-only ledger. Any modification to critical data, such as exam grades, is recorded with a precise timestamp, the exact delta of the change, and the ID of the administrative user who authorized the change. This provides absolute transparency and accountability.

8 Real-Life Aspects & Case Studies

8.1 Operational Reality at Thapar Institute

With over 30,000 students and hundreds of faculty members, TIET operates at the scale of a small municipality. The logistical nightmare of organizing mid-semester examinations involves seating matrices spanning dozens of halls, avoiding overlapping schedules for students taking multidisciplinary electives, and tracking invigilator assignments.

8.2 Case Study: The "Elective Rush" Stress Test

Historically, when elective course registration opens, the legacy portal frequently crashes or allows over-enrollment due to dirty reads. In our simulated stress tests, UniCore processed 15,000 concurrent enrollment requests in under 12 seconds. By relying on PostgreSQL's robust lock manager and maintaining a `READ COMMITTED` isolation level with targeted `UPSERT` operations, the system experienced zero deadlock rollbacks and 100% capacity enforcement.

9 Ethical Implications & Inclusivity

A centralized OS of this magnitude wields immense power. Proper management of this data requires strict adherence to ethical principles, unbiased algorithmic execution, and deep respect for the diverse ethnicity and demographics of the student body.

9.1 Inclusivity and Ethnicity Management

UniCore is designed to serve a diverse, pan-Indian, and international student demographic.

- **Algorithmic Fairness:** Allocation algorithms (such as hostel room assignments) are designed to be completely blind to ethnicity, religion, or linguistic background, relying strictly on randomized academic metrics or temporal queues (first-come, first-served).
- **Demographic Representation:** While demographic data is collected for statistical reporting to government accreditation bodies (e.g., UGC, AICTE), this data is physically partitioned and restricted via Role-Based Access Control (RBAC). It cannot be queried during standard operational transactions, ensuring zero implicit bias in system operations.
- **Accessibility Compliance (A11y):** The UniCore Frontend is designed following WCAG 2.1 AA standards, ensuring that visually impaired or differently-abled students can navigate the platform utilizing screen readers and keyboard-only interfaces. Tailwind CSS aids in enforcing strict color contrast ratios.

9.2 Data Privacy and GDPR Alignment

- **Data Minimization:** Only absolutely necessary personal identifiable information (PII) is stored.

- **Cryptographic Hashing:** Passwords and highly sensitive tokens are hashed using bcrypt with adaptive cost factors.
- **Right to Erasure / Archival:** Upon graduation, a student's operational data is archived into read-only cold storage, severing links to active operational tables.

10 Management of Work and Resources

Proper management of work is critical to delivering a platform of this complexity. The project utilizes Agile methodologies specifically tailored for full-stack infrastructure.

10.1 Tech Stack Matrix

Domain	Technologies	Purpose
Kernel / Database	PostgreSQL, BCNF	ACID Transactions, Concurrency, Locking
Backend APIs	Node.js, Express, JWT	System Call Routing, Auth, Microservices
Frontend Shell	React, Vite, Tailwind CSS	User Interface, State Management, A11y
Deployment	GitHub Actions, Docker	CI/CD Pipelines, Load Balancing

10.2 Detailed Development Timeline

The project spans a rigorous 12-week schedule.

Sprint Milestones

Phase 1: Kernel Architecture

Finalization of the 8-Domain ER Diagrams, mapping functional dependencies, and generating the core PostgreSQL Data Definition Language (DDL) scripts.

Phase 2: The Concurrency Layer

Implementation of `SELECT FOR UPDATE`, advisory locks, and PL/SQL `AFTER/BEFORE` triggers. Initial stress testing of the database utilizing pgbench.

Phase 3: API & CI/CD DevOps

Building the Node.js REST API. Implementing JWT-based Role-Based Access Control (RBAC). Setting up the GitHub Actions deploy.yml pipeline.

Phase 4: UI, UX, Web Design, and Simulation

Finalizing the React/Tailwind interfaces. Conducting end-to-end "stampede" simulations mimicking the 30,000+ student enrollment rush. Final compilation of the Technical Master Report.

11 Evaluation Metrics & Benchmarking

To objectively measure the success of the UniCore platform, we have defined strict quantitative metrics that the system must satisfy before deployment.

11.1 Performance Metrics

- **Peak Transaction Throughput (TPS):** The system must sustain a minimum of 2,500 Transactions Per Second during the simulated "Elective Rush" scenario without degrading API response times beyond 200ms.
- **Latency Percentiles:** The p95 latency for standard queries must remain below 50ms, and the p99 latency must not exceed 120ms under continuous load.

11.2 Operational Metrics

- **Time-To-Acknowledgement (TTA):** Administrative bottlenecks are monitored. The median time for a state change (e.g., clearing a library fine) to reflect across all domains must be instantaneous, enabling staff to acknowledge student requests in ≤ 2 hours.

12 Risk Management & Feasibility Study

12.1 Risk Mitigation

Risk Factor	Potential Impact	Mitigation Strategy
Database Deadlocks	System halts during complex multi-table updates.	Enforce strict lock ordering across all PL/SQL procedures.
Data Migration Loss	Corrupting data when moving from legacy silos.	Extensive use of isolated staging environments and dual-writes.
Unauthorized Access	Breaches leading to exposure of student PII.	Implementing mandatory MFA, JWT expiration, and IP whitelisting.

13 Conclusion

The UniCore Unified Campus Infrastructure represents a critical evolutionary leap for institutional infrastructure. By rigorously applying database normalization theory (BCNF) and advanced relational concurrency controls (`SELECT FOR UPDATE`, PL/SQL triggers) at the Kernel level, UniCore eliminates the endemic issues of data fragmentation and race conditions.

Moving up the stack, the robust Node.js backend ensures secure, stateless API routing, while the highly dynamic React and Tailwind CSS frontend delivers an uncompromising, accessible user experience. The entire ecosystem is bound together by automated CI/CD deployment pipelines, ensuring continuous delivery and operational resilience. Furthermore, this proposal highlights that UniCore is not just a technical endeavor; it is an ethical one. Through unbiased

algorithmic structures and stringent data privacy controls, UniCore establishes a framework that is both highly efficient and deeply equitable. The implementation of UniCore will secure the operational future of the Thapar Institute of Engineering & Technology, serving its 30,000+ students with unprecedented speed, transparency, and reliability.

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